Experiment 4
DSO and Step response of RC circuit

Components/Instruments/Meters

- Digital storage oscilloscope (DSO)
- Function generator
- Breadboard
- Resistor
- Capacitor
- Connecting wires

Part 1: Digital Storage Oscilloscope

The Digital Storage Oscilloscope (DSO) is an extremely useful and versatile laboratory instrument useful for measurement and analysis of waveforms and other phenomena in electrical and electronic circuits. An oscilloscope automatically graphs a time varying voltage, that is, it displays the instantaneous amplitude of any voltage waveform versus time. Most applications for oscilloscopes are to plot periodic signals. However, the DSO can capture (store and display) transient signals as well, with appropriate triggering.

Basic operation of the DSO

1. Screen: The DSO screen typically has 10 units on the x-axis and 8 units along the y-axis.
2. CH1 and CH2: The DSO can display signals from two different sources at the same time. The two signals can be connected to channel 1 (CH1) and channel 2 (CH2).
3. If the CH1 (and/or CH2) trace on the screen keeps getting drawn on top of itself, the waveform will be readable to the human eye. A transient signal will instantaneously appear on the screen and then disappear; it will not be recognizable to the human eye. As such, the oscilloscope is geared towards the display of periodic signals.
4. Vertical knobs for CH1 and CH2: There are two knobs right above CH1 and CH2 controlling the respective y-axis scales. Turning the knob can set the unit size on the y-axis to quantities like 5mV/div, 5V/div, etc. The two channels can have two different scales; therefore, there are two knobs for vertical axis control.
5. There is a third knob that controls the horizontal scale. Turning this knob can set the x-axis scale to quantities like 1μs/div, 1s/div, etc. The horizontal scale is common to both channels.

6. There are small knobs to control position: vertical position of CH1, vertical position of CH2, and horizontal position of both signals. These are only to shift the position of zero (never shown) on the screen.

7. There is a fourth small knob to control the TRIGGER. The concept of trigger is like that in a gun. The oscilloscope starts plotting the waveforms (CH1, CH2) when the gun fires. However, if the gun does not fire, nothing comes on the screen. The trigger can be set to fire when the periodic signal on the screen reaches a certain value.

8. The TRIGGER button controls the mode of the trigger. The trigger could be fired under different conditions; on a rising signal when it reaches a certain value, or on a falling signal when it reaches a certain value. To observe a periodic signal, the oscilloscope must be triggered properly. If the oscilloscope runs without a trigger, the (periodic) waveform might not be drawn on top of itself, and as such, the waveform will appear to be moving on the screen. Taking measurements under these conditions is not a good idea.

9. The oscilloscope can also display the CH1 signal as a function of the CH2 signal, as opposed to CH1 and CH2 as a function of time. This can be started by pressing the X-Y button. Toggling it can change the setting back to normal.

10. For manual measurements, use the CURSORS button under the Measure block. Use the circular knob to navigate along the “x” and “y” axes. Press the button to open the menu to select the x or y axis.

Voltage and frequency measurement

1. Connect the function generator to the DSO’s channel 1.

2. In the function generator, set the signal to a sine wave of 1 V**pp** and 1 kHz.

3. Measure its peak-to-peak voltage, average voltage, RMS voltage, and frequency using the DSO and tabulate the readings.

4. Measure the signal using the markings on the oscilloscope. Use the cursor to give you an accurate reading. Along with the reading, estimate an upper bound for the possible error in measurement.

5. Compare the measured values with those displayed on the function generator and those displayed on the DSO.

6. Now connect another sinusoidal signal of the same frequency to channel 2 of the DSO.

7. Can you add the two signals using the DSO? Are the results displayed consistent with the mathematically calculated values?

8. Switch to the X-Y mode on the DSO.

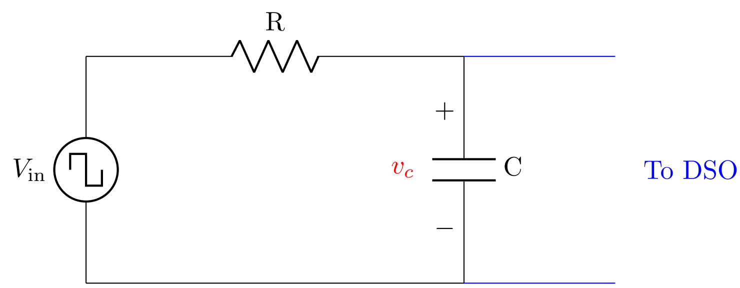
9. Set the frequency of the second signal as double the first signal’s frequency.

10. Now set the frequency of the second signal as half the first signal’s frequency.

11. Comment on your observations. Feel free to change the frequency of the input signals to figure out a pattern.

Part 2: Step response of RC circuit

1. Setup a circuit as shown below.



R	C
330 Ω, 470 Ω, 550 Ω	220 nF

2. Set the signal generator to square wave signal (Peak-to-peak value = 5 V, frequency = 50 Hz).
3. Connect 'Channel-1' of DSO to input terminals of the circuit (First terminal of "R" and second terminal of "C").
4. Connect 'Channel-2' of DSO to output terminals of the circuit (First terminal of "C" and second terminal of "C"), i.e., voltage across capacitor "C" which is v_c.
5. From the above-captured waveform, measure time constant of the circuit (Use DSO → Cursor → settings to compute the time as well as the corresponding magnitude of capacitor voltage).

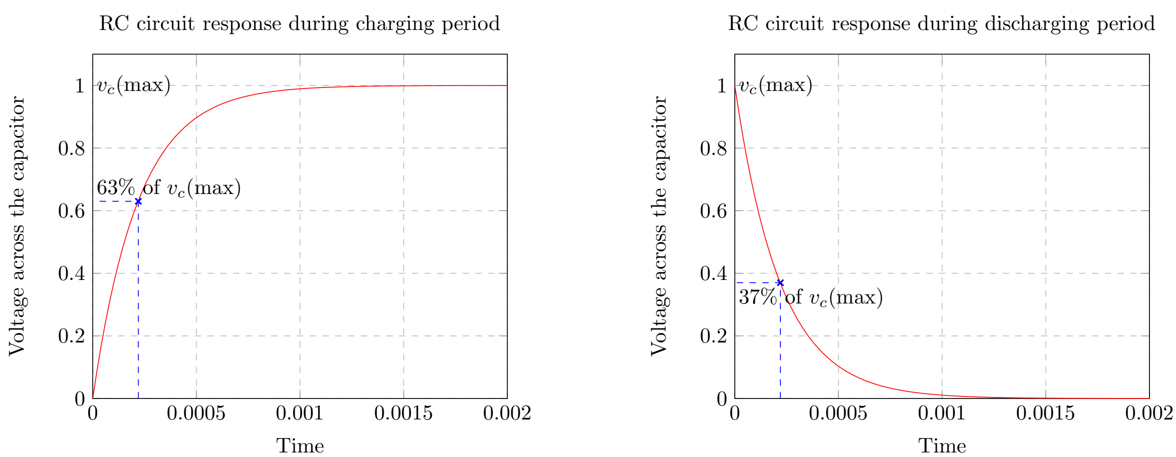


Figure 1: Typical response of a RC circuit to a square wave input (exact numbers will vary for different test cases).

6. Using mathematical formulations, compute the time constant and corresponding voltage and then compare with the measured values.

7. Use different combinations for “R” and make a tabular column of observations. Discuss the reasons for the deviation in the computed and measured values of responses.

8. Take R=1 kΩ. Adjust the input to make the voltage across the resistor in the DSO look similar to the plot for voltage across the resistor from Experiment 3 in MATLAB. Otherwise, explain why it cannot be done.